Impartial Combinatorial Game Theory

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1 Introduction

Combinatorial Game Theory is a branch of combinatorial mathematics dedicated to the mathematical analysis of games. Although originally a more recreational field of mathematics, it has developed into an actively researched area of mathematics with many open problems (Nowakowski 2017). Some of the applications of the field are algorithmic complexity, graph theory and surreal numbers.

A *combinatorial game* is one which is played without chance- players have all information about the state of the game and know every possible move they or their opponent can make at any time. An *impartial game* is a two-player game, where there is no difference between the players apart from who starts first. Each player has the same moves available in any position, and those moves have the same value. Examples of impartial games include subtraction games, Kayles and Wythoff's game. Games such as Chess and Checkers are known as *Partisan Games*, which have distinct moves for each player depending on who started first- in chess, white can move a pawn, and black can respond by moving their Knight, two distinct moves with different values. Additionally, white cannot move any of black's pieces and vice versa, so the moves available to each player depend on who's turn it is currently. Comparatively, in impartial games such as subtraction games, where players take turns subtracting numbers from a total to reach zero, if either player reaches a specific number the options available to either player will not change based on which player landed on there. They both have the same sets of moves available from that number.

One of the most influential impartial games of combinatorial game theory is Nim. Since being solved and analysed by C.L. Bouton in 1901 (Bouton 1901), Nim helped lay the groundwork for the analysis of a great many impartial games through the Sprague-Grundy theorem, which was proven by Roland Sprague in 1936 (Sprague 1935) and Patrick Michael Grundy in 1939 (Grundy 1939). Specifically, Nim has been around for centuries, the earliest references dating back to the 16th century. A game of Nim is played on piles of objects, where each player takes turns taking either one or more objects from any pile. There are no rules about the starting configuration of the piles. Generally, the player who makes the final move is the loser- but we will explore the version where the final player is the winner in this paper.

Another interesting game is Wythoff's game, a Nim variation played on 2 piles of coins, where players taking turns removing coins from either one or both piles. When taking from both piles, the number taken must be equal. Usually, the player who makes the final move is the winner. The game was historically played in China under the name "*Picking Stones*" and was solved by

Willem Abraham Wythoff in 1907 (Wythoff 1907). The solution of the game relates to Fibonacci sequences and the Golden ratio.

In this paper, we will explore the proof of Nim, expanding on it and its implications. We will also explore and prove the Sprague-Grundy Theorem for impartial games. Additionally, we will look at a variant of Nim, known as Wythoff's Game, and exploring its intricacies.

2 Defining Impartial Games

2.1 Winning and Losing Positions

We begin by defining what a game G is.

Definition 2.1. *A game G is denoted by both the position in which the game is currently on, and every position a game can move to. It is played between two players.*

We refer to a position as a specific state of a game that a player will move from, or the previous player moved to.

An Impartial Game G is defined by the moves available to both players being the same. So Chess and Checkers, where players can only move their own colour pieces, are not impartial games, whereas games like Chomp and Nim are impartial.

Definition 2.2. *A game G can be played under one of two play conventions that determine the winner:*

Normal Play defines a game such that the final player who makes a move is the winner. Equivalently, the play unable to move is the loser.

Misère Play defines a game such that the final player to move is the loser. Equivalently, the player unable to move is the winner.
(Siegel 2013)

Definition 2.3. *Let G be an impartial game, with some current position A relative to the game. We can denote G by the current position, and the possible moves which can be made. That is, $G = A = \{a_1, a_2, ..., a_n\}$, where $\{a_1, ...a_n\}$ are the positions that G can move to. Each a_i represents a position and a new possible game, denoted G', with their own set of possible moves.*

We denote the final, terminal position as $G = * = \{\emptyset\}$ where neither player can make a move. Specifically, $*$ represents the final position by the rules of that game, while $\{\emptyset\}$ represents that no move can be made from said position. The game will end once a player reaches $*$, the play convention determining the winner.

A game position can be either a $\mathcal{P}$ position or a $\mathcal{N}$ position, where:

- *A $\mathcal{P}$ position denotes the set of positions of G where the previous player that moved can force and guarantee a win.*
- *An $\mathcal{N}$ position denotes the set of positions of G where the next player, (i.e. the player who is about to play from the position), can force and guarantee a win.*

(Siegel 2013)

Note: We will refer to *Optimal Play* as a player playing without error, forcing and guaranteeing themselves a win when possible.

We see from this definition, that a game G and its sub-games, are defined recursively, by each of the moves available to a player, from each position they reach. Furthermore, we can see from the above definition, that if either player can consistently move from $\mathcal{N}$ to $\mathcal{P}$ positions, they can guarantee they win, as the opposing player will have to move from a $\mathcal{P}$ position. By the definition of $\mathcal{P}$ positions then, the previous player will win. Therefore, moving from $\mathcal{P}$ positions to $\mathcal{N}$ positions can be considered losing positions, and moving from $\mathcal{N}$ to $\mathcal{P}$ positions can be considered winning positions. We will see why moving from $\mathcal{P}$ to $\mathcal{P}$ positions and $\mathcal{N}$ to $\mathcal{N}$ positions is by definition impossible in the next section.

We can also now start to think about $\mathcal{N}$ and $\mathcal{P}$ positions, relative to normal play convention, where the final player to move is the winner, or equivalently, the player who is no longer able to make a move is the loser.

Specifically, we know that under normal play the position $G = * = \{\emptyset\}$, the terminal position, is a $\mathcal{P}$ position, as the previous player would have made the final move and won. Equivalently, the previous position, where $G = \{*, a_1, a_2, .., a_n\}$, where each a_i is a different possible position, is an $\mathcal{N}$ position, as $* = G = \{\emptyset\}$ can be reached, so the player who takes their turn in that position is the winner, given they play optimally.

2.2 Winning Impartial Games

Definition 2.4. *Let G be an impartial game. A Winning Strategy for G is a method of play that allows either the first or second player to consistently move from $\mathcal{N}$ positions and forces the other player to move from $\mathcal{P}$ positions.*

We can see from this definition, when playing a winning strategy, $\mathcal{P}$ and $\mathcal{N}$ positions will alternate with each move- for a winning strategy to work, one player must from an $\mathcal{N}$ position to a $\mathcal{P}$ position, and the other from a $\mathcal{P}$ to an $\mathcal{N}$. The following lemma proves this, giving insight into identifying $\mathcal{P}$ and $\mathcal{N}$ positions.

Lemma 2.1. *Let G be a game, played from some position A. Additionally, let $G = \{a_1, ..., a_n\}$, where each a_i is some position that G can move to, and a sub-game can be played from. We define two cases for the moves available for G:*

1. *$\forall\ a_i \in G$, $a_i \in \mathcal{N}$ $\iff$ position A is a $\mathcal{P}$ position.*
2. *$\exists\ a_i \in G$, such that, $a_i \in \mathcal{P}$ $\iff$ position A is an $\mathcal{N}$ position.*

Proof. First, we prove the forward implication:

1. For G from a position A, where for every position $a_i \in G$, $a_i \in \mathcal{N}$, then regardless of what move the current player makes from G, by the definition of an $\mathcal{N}$ position, the opposing player will be able to force a win, making A a $\mathcal{P}$ position.
2. For some game G from a position A, if there is some position $a_i \in G$, where $a_i \in \mathcal{P}$, then the player who is currently playing can move to the $\mathcal{P}$ position, forcing the player who moves after, to move from a $\mathcal{P}$ position. This means the player who is currently playing, can force a win, by the definition of a $\mathcal{P}$ position.

The inverse of both of these statements is the other. Specifically, for the inverse of statement (1), a $\mathcal{P}$ position cannot lead to another $\mathcal{P}$ position, as it would create a contradiction by the forward implication of (2), so every position of a $\mathcal{P}$ position must lead to a $\mathcal{N}$ position.

For (2), an $\mathcal{N}$ position must lead to a $\mathcal{P}$ position, as otherwise it would only lead to $\mathcal{N}$ positions, creating a contradiction by the forward implication of (1), so every $\mathcal{N}$ position must lead to a $\mathcal{P}$ position. □

Remark. *This is how a win is forced by one of the players- by continually forcing the other player to move from $\mathcal{P}$ positions to $\mathcal{N}$ positions that they can move from $\mathcal{N}$ positions to $\mathcal{P}$ positions.*

Next, we explore the proof of a subtraction game- specifically ***21***. The rules of ***21*** are that each player takes turns adding either 1, 2 or 3, starting from 0, the loser being the person who lands on 21. (Note that turns cannot be skipped) However, we will look at a slightly different, generalised version, where two players begin from some $k \in \mathbb{N}$ and count down, taking turns subtracting some $i \in \{1, 2, ...n\}$. We will also be looking at the game under normal play convention version, so the winner is the player who makes the final move to 0.

Theorem 2.1. *Let G be a subtraction game under normal play convention with similar rules to* ***21****. The game starts on some position $A = k \in \mathbb{N}$, and each player takes turns subtracting some number $i \in \{1, 2, ...n\}$. Then $\forall$ position $A \in \{0, 1, 2, ..., k\}$,*

$$A \mod (n+1) \equiv 0, \iff A \in \mathcal{N}. \tag{1}$$

and

$$A \mod (n+1) \not\equiv 0, \iff A \in \mathcal{P}. \tag{2}$$

Proof. Let $G = \{a_1, a_2, ..., a_m\}$ denote a game, where each $a_l \in G$ is a possible move for a player to a new sub-game G'.

Since we are playing under normal play convention, the terminal position (which we denoted generally as $*$), will be $0 = G = \{\emptyset\}$, and is a $\mathcal{P}$ position. Then, any position from which $0 \in G$, is an $\mathcal{N}$ position by Lemma 2.1. We know that $0 \in G$, for any G from position $0 + i$, making the positions $\{1, ..., n\} \in \mathcal{N}$. Therefore, by Lemma 2.1, the position $n + 1$, from which $G = \{1, ..., n\}$, is a $\mathcal{P}$ position, since every $a_i \in G$ is an $\mathcal{N}$ position.

From the position of $n+1$ then, using Lemma 2.1, we see that the positions $\{n+2, ..., 2n+1\}$ are all $\mathcal{N}$ positions, as for these positions, the position $(n+1) \in G$. Furthermore, $2n+2$ is a $\mathcal{P}$ position, since from this position, $G = \{n+2, ..., 2n+2\}$, all of which are $\mathcal{N}$ positions.

We start to see a pattern- for all $A \in \mathcal{P}$, $A \mod (n+1) \equiv 0$, since every position, $x(n+1)$, where $x \in \mathbb{N}_0$, can only reach $\mathcal{N}$ positions, as you can only subtract at most $i = n$, preventing you from reaching the next $\mathcal{P}$ position. Equivalently, since every other position where $A \mod (n+1) \not\equiv 0$, can reach a $\mathcal{P}$ position (as they are at most only n positions away), they are $\mathcal{N}$ positions.

The inverse of claim (1) and (2), is the inverse of each other statement, by Lemma 2.1. We can conclude through induction on the base case of position $0 \in \mathcal{P}$ and forming an inductive hypothesis using Lemma 2.1, that (1) and (2) are true. □

We now know the general strategy for creating a solution/optimal strategy to an impartial game- identifying the $\mathcal{P}$ and $\mathcal{N}$ positions and then identifying a strategy for either the first or the second player to consistently move from $\mathcal{N}$ positions. Next, we will give an explanation and short example game of Nim, along with some notation.

3 Nim

3.1 Definition

Definition 3.1. *Nim is game of picking counters from piles. Specifically, we start on a position of n piles, denoted as* $(x_1, ..., x_n)$*, where each* $x_i \in \mathbb{N}$*, represents the size of pile i. (Starting piles cannot be of size 0)*

Each player must make a move by taking one or more counters from a single pile, up to taking a whole single pile. The game ends once all counter piles are empty, making $G = *$ *the position where each pile has a value of 0.*

Example 3.1. *Let G be a game of Nim under normal play convention, played on 2 piles of counters of height 2, 3 respectively. We denote each position in G as* (x_1, x_2)*. The terminal position* $*$ *is* $(0, 0)$*, that is,* $G = (0, 0)$*, denoting the end of the game and that the set of possible moves is* $\{\emptyset\}$*.*

The initial position is $(2, 3)$*. We see* $G = (2, 3) = \{(0, 3), (1, 3), (2, 2), (2, 1), (2, 0)\}$*.*

Player 1 makes a move, removing 2 counters from x_2*. We see* $G = (2, 1) = \{(0, 1), (1, 1), (2, 0)\}$*.*

Player 2, makes a move, removing a counter from x_1*. We see* $G = (1, 1) = \{(1, 0), (0, 1)\}$*.*

Player 1 moves again, taking a counter from x_1*. We see* $G = (0, 1) = \{(0, 0)\}$*.*

Player 2 moves, and with only a single move left to take a counter from the second pile, they win the game. It reaches the terminal position of $G = (0, 0) = \{\emptyset\}$*- so no other move can be made.*

We will next start to discuss the analysis of Nim, eventually looking into how we can identify $\mathcal{P}$ and $\mathcal{N}$ positions.

Definition 3.2. *The Mirror Image Strategy, is a strategy for play where the second player mimics to first player's each move.*

With this strategy we see that if we have a game of Nim with (x, x), 2 piles of $x \in \mathbb{N}$ counters, it is a $\mathcal{P}$ position. Any move the first player makes in either pile, can simply be copied by the second player during their turn on the other pile. Eventually, the first player will have to make a pile 0, meaning the second player can win by taking all the other pile.

Definition 3.3 (Nim-Sum)**.** *Let G be a game of Nim under normal play convention. We denote Nim-Sum, as* $\oplus$*. It is the XOR (exclusive-or) operation that is performed on binary numbers. It is performed by converting each Nim pile into its binary counterpart, and then "XORing" them by adding them in mod 2, without "carrying" over. Specifically, we can define this with the following, quoted equation:*

"Let $x, y \in \mathbb{N}$*, be Nim piles. They are expressed in Base 2 as* $x = \sum_{i=0}^{n} x_i 2^i$ *and* $y = \sum_{i=0}^{n} y_i 2^i$*, with* $x_i, y_i \in \{0, 1\}$*. Nim-sum is then defined as:*

$$x \oplus y = \sum_{i=0}^{n} w_i 2^i$$

where $w_i = x_i + y_i \pmod 2$*." (Miyadera and Manabe 2022)*

In simpler terms, Nim-Sum is binary addition where the addition for each 1 and 0 will be done according to Table 1.

$\oplus$	1	0
1	0	1
0	1	0

Table 1: XOR Addition

Nim-Sum gives the number of "extra" counters in binary. Since binary has a unique representation of every $\mathbb{N}$ number, calculating the Nim-sum of all Nim piles in a game gives us the binary "bits" without matching pairs in the binary of other Nim piles. A Nim-Sum of zero, therefore tells us that all piles are equal, with all of them having corresponding binary "bits" in other piles. We can therefore use Nim-sum to give us a picture about the overall state of a game of Nim.

We denote the Nim-sum of games as **Nimbers**. These allow us to model a game of Nim as a single pile, a nimber of size 0 implying an empty "pile", and an equal nim-sum of the overall game. We denote a nimber as $*x$, where $x \in \mathbb{N}$ could be the nim-sum of a game, or the nimber corresponding to a specific pile.

Lemma 3.1. *The operation of Nim-Sum forms an Abelian group under* $\mathbb{N}_0$*, where* $\mathbb{N}_0 = \{\mathbb{N} \cup \{0\}$*. Specifically,* $(\mathbb{N}_0, \oplus)$

Proof.

- **Closure** $\forall\ a, b \in \mathbb{N}$, $a \oplus b \in \mathbb{N}_0$.
  Since XOR is a binary operation, and the result of XOR gives another binary number, which can be converted to a natural number.

- **Associativity** $\forall\ a, b \in \mathbb{N}_0$, $(a \oplus b) \oplus c = a \oplus (b \oplus c)$.
  XOR is associative in the binary system, so this operation also holds. The order in which the ones or zeroes are "XOR-ed" does not change the outcome.

- **Existence of Identity Element** $\forall\ a \in \mathbb{N}_0$, $a \oplus 0 = a$.
  The identity element is therefore 0, since a binary sequence of all zeroes cannot change a one to a zero.

- **Existence of Inverse** $\forall\ a \in \mathbb{N}_0$, $a^- = a$.
  The inverse of a number under Nim-sum is the number itself.

- **Commutativity** $\forall\ a, b \in \mathbb{N}_0$, $a \oplus b = b \oplus a$.
  XOR is trivially commutative since each digit in the first or second binary numbers is compared separately, and the order of digits does not change the result, as can be seen by table 1. Therefore, Nim-sum is also commutative.

□

Since nim-sum gives us the state of a game of Nim in a single number, which is equivalent to having a single pile of the same size. We denote this new single pile as the *Nimber* of the current Nim position. It is also important to note that a single nim pile can never have a nim-sum of *0, unless it is empty, since it is a single binary number.

3.2 Proof of Nim

We will now explore a slightly adapted version of the proof of Nim by C.L. Bouton. (Bouton 1901).

Theorem 3.1. *Let G be a game of normal play Nim with position A. G is a $\mathcal{P}$ position if the Nim-sum of G is zero. Otherwise, it is a $\mathcal{N}$ position.*

Proof. We will use proof by induction with 2 cases- moving from nim-sum zero will always guarantee that we go to a non-zero position, and moving from a non-zero position always guarantees a move to a zero position.

Base case: Let the position of $G = (a_1, ..., a_n)$, where every a_i, is an empty pile. Then we see that $a_1 \oplus a_2 \oplus ... \oplus a_n = 0$. This is the terminal position, and therefore a $\mathcal{P}$ position.

Case 1 (zero to non-zero): Let G be the previous position of the game, and let G' be the position of the game after a move has been made. G has position $A = (a_1, ..., a_n)$ and G' has position $B = (b_1, ..., b_n)$, where each $a_i, b_i \in \mathbb{N}$ are singular piles, and each b_i represents each a_i after the move has been made. Let $*a_i$ and $*b_i$ denote the nimber values/binary value of each individual pile and let $a_1 \oplus a_2 \oplus ... \oplus a_n = 0$.

Let $t = (a_1 \oplus ... \oplus a_n) \oplus (b_1 \oplus ... \oplus b_n)$. By the rules of Nim, after a move has been made, some single pile $a_l \in A$, must have been reduced by some number x, to a new pile $b_l \in B$. Since it is only a single pile, every other $a_i = b_i$. (i.e. Every pile is the same, before and after, except some pile a_l and b_l). Next, we use the inverse defined in Lemma 3.1 (that the inverse of a nimber is itself) to prove the following, where $*a_i$ is the binary/nimber counterpart of each a_i:

$$\begin{aligned} t &= (*a_1 \oplus *a_2 \oplus ... \oplus *a_n) \oplus (*b_1 \oplus *b_2 \oplus ... \oplus *b_n) \\ &= (*a_1 \oplus *b_1) \oplus (*a_2 \oplus *b_2) \oplus ... \oplus (*a_i \oplus *b_i) \oplus ... \oplus (*a_n \oplus *b_n) \\ &= (*0) \oplus (*0) \oplus ... \oplus (*a_i \oplus *b_i) \oplus ... \oplus (*0) \\ &= (*a_l \oplus *b_l). \end{aligned} \tag{3}$$

Hence, the nim-sum of G' cannot be $*0$, as otherwise it would imply $*a_l = *b_l$, a contradiction, since inverses are unique by Lemma 3.1 and because $b_l = a_l - x$.

Case 2 (non-zero to zero): Let G have position $A = (a_1, ..., a_n)$ and let G' have position $B = (b_1, ..., b_n)$. Let t be the nim-sum of all $a_i \in A$, and $t \neq 0$, and let s be the nim-sum of all $b_i \in B$. (So t is the nim-sum of A, before the move has been made, and s is the nim-sum after the move was made)

We need to define a move in G, such that $s = 0$. Consider the binary representation of t. We select some pile a_m to subtract from, that is greater than or equal to the largest power of 2 in t (so the bit farthest to the left). We know such a pile exists, as otherwise t itself would not contain such a binary bit. We calculate and perform the move $b_m = a_m \oplus t$, where b_m will be the pile a_m after making the move. Doing this removes the largest offset bit of t, reducing the pile so that it only contains the correct bits to balance t, making $s = 0$ once the move is played.

For example, if $t = 1011$, The presence of 1 in 2^3, implies at some pile, suppose a_m, has a 2^3 bit. Therefore, we can reduce some pile a_m to match $a_m \oplus t$. After which, the 1 in 2^3 disappears from t, with the b_m being such that $s = 0$, since we calculated it by $a_m \oplus t$.

With this we can conclude the proof, since every position A where the nim-sum of the game is not zero, can make a move to make the game nim-sum zero. Then from a position of nim-sum zero, the player must make the nim-sum non-zero again, as they can only change a single pile. Through induction then, the nim-sum can continually be reduced to zero and the game will eventually reach $G = * = \{\emptyset\}$. Hence positions with nim-sum of 0 are $\mathcal{P}$ positions, and nim-sum $\neq 0$ positions are $\mathcal{N}$ positions. (Bouton 1901) □

4 Sprague-Grundy Theorem

4.1 Nimbers and Mex

In the Nim subsection, we stated that nim-sum can relate the state of a game of Nim to a single pile, giving us an equivalent nimber for that game. We now want to try to generalise the concept of nimbers to all impartial games, through using nim-sum, which allows for a consistent method to find solutions for impartial games and further analysis. We will generalise calculating nimbers through the concept of *mex* and by proving that greater nimber values can be ignored, through the proof of *Poker Nim*. It is important to note that we can find the nimber of any impartial game position by calculating the nim-sum of the nimbers of all possible moves from that position. We will denote the nimber of a specific game position as $*m$, for $m \in \mathbb{N}_0$.

We now discuss Poker Nim - a variation of Nim where either player, instead of taking from a pile, can add any number to any single pile.

Lemma 4.1. *(Poker Nim) Regardless of if a player decides to add counters, as long as both players are playing optimally, it does not change $\mathcal{P}$ and $\mathcal{N}$ positions.*

Proof. We first see that if a player is on an $\mathcal{N}$ position, there is no need to add any number to any pile, as the new position will be either an $\mathcal{N}$ or $\mathcal{P}$ position. The same as if they were to move normally, meaning it is better to move towards to end of the game and remove a number from some pile and go to a $\mathcal{P}$ position.

We next see that if a player is on a $\mathcal{P}$ position, then if they add to any pile, the move can simply be reversed by the other player to take them back to the original position. Then if they remove any counters the game will be played as normal. (Berlekamp 2001) □

Definition 4.1. *Let $S \subset \mathbb{N}_0$. We define the **mex** of S (minimal excluded value) as the smallest integer that is not in S. i.e.*

$$mex(S) = \min\{\mathbb{N}_0 \setminus S\}.$$

Note: $mex\{\emptyset\} = 0$. (Siegel 2013)

Theorem 4.1. *Mex (Minimum Excludant) Rule*

Let $G = \{a_1, a_2, ..., a_n\}$ be a game of Nim on some position A. Let the nimber $*A$ be the nim-sum calculated for A. Let $*a_i$ be the nimber value of each $a_i \in G$, calculated through nim-sum. Then $*A = *m$, where $*m = mex\{*a_1, *a_2, ..., *a_n\}$

Proof. We can prove this by looking at the new positions available from G. Specifically, for $*a_i < *m$, we see that all such moves possible from $*m$, are also possible from $*A$. We see that these $*a_i$ are consecutive integers, similar to a single Nim pile. We also see that moving to higher nimber positions is useless by Lemma 4.1. Additionally, since $*m \notin G$, it is an unreachable position from G, meaning that it will be consistent for any position A. Hence, $*A = *m$, since $*A$ acts the same as a nim pile of size $*m$. (Berlekamp 2001) □

We can define the nimber of any position in any impartial game now, by induction of the base case of $G = mex\{\emptyset\} = *0$. Specifically, by using mex, we can define positions where $G = \{*0\}$ (i.e. the only move is to the terminal position) as having nimber value of $*1$. For positions that go to either $*1$ or $*$ ($G = \{*, *1\}$), they have value $*2$, and so on. When a game exists without $*0$ as a position, it is given value $*0$ as well, representing that any move from this position has a non-zero nimber. We explore a formal proof of this in the next section.

4.2 Nimbers in Impartial Games

Recall that we defined impartial games as games where the moves available to each player are only dependant on the position a player is moving from, and not dependent on which player is moving. We will now explore how we can use the concept of nimbers and nim piles to generalise impartial games using a theorem by Roland Sprague (Sprague 1935) and Patrick Michael Grundy (Grundy 1939).

Theorem 4.2. *(The Sprague-Grundy theorem) Let G be any impartial game. Then $G = *m$ for some $m \in \mathbb{N}_0$.*

Proof. For G we apply induction from the base case of $G = * = \{\emptyset\}$, i.e. the game is in the terminal position and has no moves. Recall the $mex\{\emptyset\} = *0$. Any position then, which leads to $*$ then has value of $*1$, since $mex\{*0\} = *1$. Continuing this, we see that $mex\{0*, *1\} = *2$. Then, the first position we see that has $G = \{*a_1, *a_2 ... * a_n\}$, where every every position $a_i \neq *0$, has nimber $*0$, that is $G = *0$. By this reasoning we can find the nimber of any position, determining the nimber of any impartial game and its position. Hence G can be modelled as a game of Nim using Theorem 4.1, with the current nimber and position being $*m = mex\{*a_1, *a_2, ... * a_n\}$. (Grundy 1939) (Sprague 1935) (Siegel 2013) □

The consequences of this theorem are quite significant, since it allows for the calculation of the nature of any position in any impartial game, through finding the first unavailable nimber position, using it to model a game of Nim. Then, by finding a move which takes the nimber position to $*0$, we can form a strategy to win. We can guarantee such a strategy exists through the proof of Nim in Theorem 3.1. Specifically, Theorem 3.1 proves that there is always a move to nimber $*0$, and through induction, a winning strategy exists. We can use Lemma 2.1 to assign $\mathcal{P}$ and $\mathcal{N}$ to the position for which a nimber is calculated from. Note that we also refer to the nimber calculated this way, as the *Grundy* number of a position as well.

Corollary 4.1. *A position with Grundy number of $*0$ is a $\mathcal{P}$ position. All other positions are $\mathcal{N}$ positions.*

Proof. From the proof of Theorem 3.1, we know that a move always exists to a nim(ber) positions of $*0$. The Grundy number of a terminal position is also $*0$. Therefore, through induction and Theorem 3.1, we guarantee a winning strategy. Furthermore, with Lemma 2.1, we assign positions as being $\mathcal{P}$ or $\mathcal{N}$, positions with Grundy number/nimber $*0$ being $\mathcal{P}$ positions. □

Example 4.1. *Let G be an impartial game under the rules of 21, as explained before. We modify the games that it begins on the number 7 (for simplicity of this example).*

We calculate the nimber of each position recursively:

$$
\begin{aligned}
mex(0) =& mex(\emptyset) = *0 \\
mex(1) =& mex\{mex(0)\} = *1 \\
mex(2) =& mex\{mex(1), mex(0)\} = *2 \\
mex(3) =& mex\{mex(2), mex(1), mex(0)\} = *3 \\
mex(4) =& mex\{mex(3), mex(2), mex(1)\} = *0 \\
mex(5) =& mex\{mex(4), mex(3), mex(2)\} = *1 \\
mex(6) =& mex\{mex(5), mex(4), mex(3)\} = *2 \\
mex(7) =& mex\{mex(6), mex(5), mex(4)\} = *3.
\end{aligned}
$$

Hence, we can form the following table:

Position	0	1	2	3	4	5	6	7
Grundy Number	$*0$	$*1$	$*2$	$*3$	$*0$	$*1$	$*2$	$*3$

Table 2: Grundy Numbers/Nimbers of modified *21*

We see that the Grundy numbers are the same as what we calculated in Theorem 2.1, with the $\mathcal{P}$ positions being on the positions with Grundy number of 0, and all other positions being $\mathcal{N}$ positions.

5 Wythoff's Game

5.1 Winning Wythoff's Game

Next, we'll discuss a variation of Nim, known as Wythoff's Nim or Wythoff's game- named after the mathematician who solved it, Willem Abraham Wythoff. The variation is played on 2 piles of coins/tokens, where each player takes turns taking any number from one pile (as in Nim), or taking the same number from both piles. So, a game could be with piles $(6, 7)$, where the first move could be to move to $(3, 4)$, and the second move to $(3, 3)$, and the final move to $(0, 0)$- making player 1 the winner. What is interesting about this game, is its link to the golden ratio.

Theorem 5.1. *Let G be a game of Wythoff's Game under normal play, with each position being represented by an ordered pair $(a, b) \in \mathbb{N}_0^2$. We index the $\mathcal{P}$ positions, starting from (0,0) being (a_0, b_0). The nth $\mathcal{P}$ position, (a_n, b_n), is then defined recursively by:*

$$
\begin{aligned}
a_n &= mex\{a_i, b_i : i < n\} \\
b_n &= a_n + n
\end{aligned}
$$

Note: $a_0 = mex\{\emptyset\} = 0$

Remark. *What this theorem describes is that the first pile of each $\mathcal{P}$ position, can be found by finding the mex of all previous $\mathcal{P}$ positions- i.e. the first number has not appeared in any previous $\mathcal{P}$ positions. The second pile can then be found by adding the index n of that $\mathcal{P}$ position, to the first pile.*

Proof. By Lemma 2.1 and induction on n, it suffices to show that each (a_n, b_n) have no move to any other $\mathcal{P}$ position (a_m, b_m), for $m < n$ (therefore showing they only move to $\mathcal{N}$ positions), and that every (a_n, b_n) is reachable by an $\mathcal{N}$ position.

To start off, we will prove that every $\mathcal{P}$ position leads only to $\mathcal{N}$ positions. We know that $(0, 0)$ is a $\mathcal{P}$ position, since it is the terminal position, so $(a_0, b_0) = (0, 0)$, and $(a_1, b_1) = (1, 2)$, $(a_2, b_2) = (3, 5)$... are $\mathcal{P}$ positions by the theorem.

There are 3 possible moves to make from each (a_n, b_n), removing a number from a_n, removing a number from b_n, or removing an equal number from both a_n, b_n. Since $a_n \neq b_m$ and $a_n \neq a_m$, for any $m < n$ by the definition of a_n, we can conclude that removing a counter from b_n, will never give us a move to an (a_m, b_m). We next see that $b_n \neq b_m$ and $b_n \neq a_n$. We know $a_n > a_m$, by how a_n is recursively defined by mex, which means that every $b_n > b_m$. Therefore, every $b_n \neq b_m$, and $b_n \neq a_n$, so there is no move that can be done to a_n to form (a_m, b_m). We see that for the final possible move of removing an equal number from both piles, $a_n - b_n = n$, while $a_m - b_m = m$, and since $m \neq n$ by how m is defined, we can say that there is no (a_n, b_n) that leads to another (a_m, b_m).

Next we prove that every $\mathcal{N}$ leads to a $\mathcal{P}$ position - that every (a, b) leads to some (a_n, b_n) (The inverse of Lemma 2.1). First suppose that $a = a_n$, for some n. If then $b > b_n$, we can reduce b to b_n. If $b < b_n$, we see that $a_n \leq b < b_n$. Let $m = a - b$, then $m < n$, hence we can find an $a_m < a_n$, where the pair (a_m, b_m) have difference m. Therefore, there is a move to some (a_m, b_m) by reducing both piles. Lastly, we look at if $a \neq a_n$. We see that a_n and b_n are calculated using mex, meaning every natural number will appear in some position. Therefore, $a =$some b_n, and $b \geq a > a_n$, so there is a move to (b_n, a_n), by reducing b to the matching a_n.

Therefore, one player can constantly from a $\mathcal{N}$ position, and force the other player to move from a $\mathcal{P}$ position, and by induction, we will eventually reach $G = (0, 0)$ by either reaching a point where one pile is 0, or both piles are equal. (Siegel 2013) □

An observation about the $\mathcal{P}$ positions we have defined in Theorem 5.1, is that when described on a two-dimensional graph (x-axis is a and y-axis is b), they tend to cluster around 2 lines with gradients ϕ, $1/\phi$, where $\phi = \frac{1+\sqrt{5}}{2}$, the golden ratio. Intuitively, we can see this by how in Theorem 5.1, we add the index to find the second pile b_n, similar to forming a Fibonacci sequence.

While Theorem 5.1 gives us a method for finding the $\mathcal{P}$ and $\mathcal{N}$ positions recursively, it does not give us a true method to find a specific n^{th} $\mathcal{P}$ position. To do this, we must first discuss complementary sets, irrational numbers.

Definition 5.1. *Let $\mathcal{A}, \mathcal{B} \subset \mathbb{N}$. $\mathcal{A}, \mathcal{B}$ are complementary if $\mathcal{B} = \mathbb{N} \setminus \mathcal{A}$. That is, $\mathcal{A} \bigcap \mathcal{B} = \emptyset$ and $\mathcal{A} \cup \mathcal{B} = \mathbb{N}$.*

The following proof applies to *all* irrational numbers and their reciprocals. These are specifically known as Beatty sequences, and were proven by John William Strutt Baron Rayleigh in 1877 (Rayleigh 1877) and in The American Mathematical Monthly by several other mathematicians (Beatty, Ostrowski, et al. 1927), after Samuel Beatty proposed the problem the year prior (Beatty, Altshiller-Court, et al. 1926).

Lemma 5.1. *Fix irrational real numbers $\alpha, \beta > 1$, such that:*

$$\frac{1}{\alpha} + \frac{1}{\beta} = 1.$$

Then the sets

$$\{\lfloor n\alpha \rfloor : n \in \mathbb{N}\} \text{ and } \{\lfloor n\beta \rfloor : n \in \mathbb{N}\}$$

are complementary.

Proof. Let

$$\mathcal{A} = \left\{ \frac{i}{\alpha} : i \in \mathbb{N} \right\} \text{ and } \mathcal{B} = \left\{ \frac{j}{\beta} : j \in \mathbb{N} \right\}.$$

Since $1/\alpha + 1/\beta = 1$, we can rearrange to form an irrational fraction: $\frac{\alpha}{\beta} = \alpha - 1$. Therefore, we see that $\mathcal{A} \cap \mathcal{B} = \emptyset$. Since otherwise, if there is any element in both sets:

$$\frac{i}{\alpha} = \frac{j}{\beta} \implies \frac{i}{j} = \frac{\alpha}{\beta} = \alpha - 1$$

which is a contradiction, since $i, j \in \mathbb{N}$ and cannot be irrational.

Next, we set $\mathcal{C} = \mathcal{A} \cup \mathcal{B}$. We define a function $f : \mathcal{C} \to \mathbf{N}_0$, where $f(c) = |\{d \in \mathcal{C} : d \leq c\}|$. That is, $f(c)$ gives the number of elements of $\mathcal{C}$ that are less than c. We see that since $\mathcal{A}$ and $\mathcal{B}$ are disjoint, $f[\mathcal{A}]$ and $f[\mathcal{B}]$ are complementary (since being disjoint means that for every $c, c' \in \mathcal{C}$, where $c \neq c'$, $f(c) \neq f(c')$, since c, c' are unique members of $\mathcal{A}$ or $\mathcal{B}$).

Now let $c \in \mathcal{A}$, so $x = i/\alpha$. Then

$$f(c) = |\{d \in \mathcal{A} : d \leq c\}| + |\{d \in \mathcal{B} : d \leq c\}|.$$

We see that for

$$|\{d \in \mathcal{A} : d \leq c\}| = i$$

and for

$$\begin{aligned} |\{d \in \mathcal{B} : d \leq c\}| &= |\{j \in \mathbb{N}^+ : j/\beta \leq c\}| \\ &= |\{j \in \mathbb{N}^+ : j \leq \beta c\}| \\ &= \lfloor \beta c \rfloor. \end{aligned}$$

Now we see that $c\beta = i\beta/\alpha = i(\beta - 1)$ (since $\beta/\alpha = \beta - 1$, the inverse of the state from earlier). So

$$f(c) = i + \lfloor c\beta \rfloor = i + \lfloor i(\beta - 1) \rfloor = \lfloor i\beta \rfloor.$$

We use the same argument to show that for $x \in \mathcal{B}$, we get

$$f(c) = j + \lfloor c\alpha \rfloor = j + \lfloor j(\alpha - 1) \rfloor = \lfloor j\alpha \rfloor.$$

Therefore, $\mathcal{C}$ is a disjoint union of $\mathcal{A}$ and $\mathcal{B}$. Hence, we conclude that the sets

$$\{\lfloor n\alpha \rfloor : n \in \mathbb{N}\} \text{ and } \{\lfloor n\beta \rfloor : n \in \mathbb{N}\}$$

are complementary. (Siegel 2013) □

Theorem 5.2. *(Wythoff positions)*
The nth $\mathcal{P}$ position of Wythoff's game is given by

$$(a_n, b_n) = (\lfloor n\phi \rfloor, \lfloor n\phi^2 \rfloor).$$

where $\phi = \frac{1+\sqrt{5}}{2}$.

Proof. We can easily verify that $\phi^2 = \phi + 1$, by simplifying $\phi^2 = (\frac{1+\sqrt{5}}{2})^2$. We can then take $\phi^2 = \phi + 1$, and rearrange by dividing through by ϕ^2, giving us

$$\frac{1}{\phi^2} + \frac{1}{\phi} = 1.$$

We next see that $n\phi^2 = n\phi + n$, therefore

$$\lfloor n\phi^2 \rfloor = \lfloor n\phi + n \rfloor = \lfloor n\phi \rfloor + n.$$

So, by Lemma 5.1, $\lfloor n\phi \rfloor$ and $\lfloor n\phi^2 \rfloor$ are complementary. The fact they are complementary, therefore implies that

$$\lfloor n\phi \rfloor = mex\{\lfloor i\phi \rfloor, \lfloor n\phi^2 \rfloor : i < n\} \text{ and } \lfloor n\phi^2 \rfloor = \lfloor n\phi \rfloor + n.$$

The theorem now follows from Theorem 5.1, with $a_n = \lfloor n\phi \rfloor$ and $b_n = \lfloor n\phi^2 \rfloor = \lfloor n\phi \rfloor + n$. (Siegel 2013) □

6 Future Work

From the proof of both Nim and Wythoff's game shown in this paper, it is clear there is still much to be explored in the analysis of impartial games. Despite being able to find the $\mathcal{P}$ and $\mathcal{N}$ positions, there is no known general formula for finding the Grundy numbers of Wythoff's game (Siegel 2013). Additionally, for Nim and various other impartial games, there are currently open problems for finding general formulae for the $\mathcal{P}$ and $\mathcal{N}$ positions of variations of these games, played with a pass move to skip their turn, that a singular player can use once (Nowakowski 2017). For example, in the case of Nim, we see that allowing either player to pass a single time changes how we need to view the game, looking instead at the game pre-pass and post-pass, with post-pass being the same as normal Nim. For pre-pass, we see that the game $\mathcal{P}$ and $\mathcal{N}$ positions are quite different, as players now need to avoid moving to $\mathcal{P}$ positions, as otherwise the other player can pass, causing the player originally moving to now have to take the losing move.

Other points of analysing these games is to analyse Misére variations, where the final player to move is the loser. Generally speaking, these are quite similar to the normal play versions with slightly differing $\mathcal{P}$ and $\mathcal{N}$ positions, yet similar general formulae.

Another branch of Nim is generalising it as an Octal game and further as a Heap game. Heap games consist of multiple piles or "heaps", where players take turns taking from any single pile according to the rule-set of the game. Octal games consist of removing counters from piles and then optionally splitting the tokens into two separate piles. The rules of the game specify the circumstances by which any specific number of tokens may be taken from a pile, and if the heap can split afterwards. An interesting notation of Octal games is that specific games are described by a decimal $d_0.d_1d_2d_3...$, where each $0 \leq d_i < 8$. The ith digit d_i represents the circumstances in which removing i tokens may be a legal move. That is, each number 0 through 7 represents a unique circumstance when we should remove i counters. For example, 0 represents it is never legal to remove i counters, while 1 represents that it is legal to remove i counters if there are no remaining counters after the move is made. The meaning of each code is in Table 3, as well as the binary version of each. It is also worth mentioning that Nim has the code 0.333..., and that Kayles has code 0.77 (Siegel 2013).

d_i	Binary	It is legal to remove i tokens ...
0	000	Never.
1	001	Provided that no tokens remain after they are removed.
2	010	Provided that at least one token remains (and any remaining tokens are left in a single heap).
3	011	Always (but the remaining tokens must be left in a single heap).
4	100	Provided that the remaining tokens are split into exactly two nonempty heaps.
5	101	Provided that *if* any tokens remain, *then* they are split into exactly two nonempty heaps.
6	110	Provided that at least one token remains; the remaining tokens may optionally be split into two heaps.
7	111	Always; the remaining tokens may optionally be split into two heaps

Table 3: The various octal codes (Siegel 2013).

Another topic of research that has grown in the past few decades is surreal numbers, which is an off-shoot of partizan games; however, they use Nim-sum in how the sum of surreal numbers is calculated. The system of surreal numbers describes a fully ordered class of numbers containing real numbers, infinitesimal numbers, and infinite numbers (Conway 2000).

7 Conclusion

We have explored the basics of combinatorial game theory for impartial games, specifically the fundamental concepts of $\mathcal{P}$ and $\mathcal{N}$ positions, along with applications in Nim, Wythoff's game and a generalisation of *21*. Through the proof of Nim, we demonstrated how the concept of Nim-sum determines winning and losing positions, establishing a fundamental strategy for optimal play. The Sprague-Grundy theorem further generalised these principles, allowing us to model any impartial game in terms of nimbers, thereby providing a systematic approach to determining game outcomes. Additionally, our analysis of Wythoff's Game and its connection to the golden ratio and Beatty sequences allowed us to determine a general formula for finding the $\mathcal{P}$ positions. The results discussed in this paper open discussion for various different paths of research into the complexity and solutions of impartial games, and more generally, combinatorial games.

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